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Atlantic
Technological
University

Module Title: Embedded Systems 1
Module Code: EMBS_ME701
Year: 2026-2027
Assignment: CA Assignment 1
Submission: 11:59pm Friday 6th November 2026

Instructions to Student:

Any evidence of plagiarism from another submission will result in **BOTH** submissions being marked down.

This CA will assess your understanding of weeks 1-6 of the module.

The assignment written report and commented code source files (CSS project Files) should be submitted via Blackboard in the (EMBS_701 Assessment area) by **11.59pm Fri 7th November 2025.**

Module Coordinator: Dr. John Wade.

Title: Design and Implementation of an air duct actuator system

Assessment Weight: Constitutes 20% of the overall mark

System Requirements

Using an MSP430 MSP-EXP430G2E educational kit as the target hardware, you are asked to develop a microcontroller system solution using the Code Composer Studio development environment.

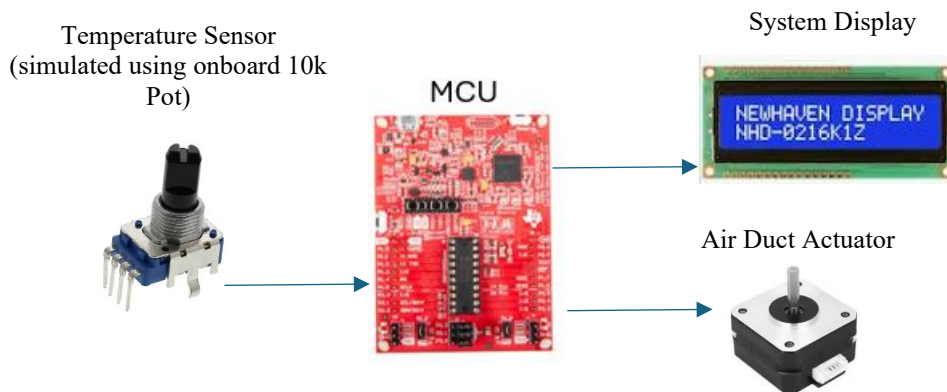


Fig.1 Desired System

The desired system (see Fig. 1) will be a prototype solution for a **temperature controlled airduct actuator systems**. The prototype system will consist of 1 MSP-EXP430G2ET Educational kit and will use the onboard Launchpad MCU development board, 10 Pot to simulate the systems temperature sensor, LCD Screen to display current temperature and the systems current state i.e. airduct open/closed, and the onboard stepper motor or a servo motor to simulate the actuator.

- The MCU can be considered the control system for the airduct system and will read the temperature and decide if the actuator should be open or closed. It will also display relevant information to the user on the onboard LCD screen. This information should include, but is not limited to, the current measure temperature and the state of the actuator (open/closed).
- Basic System Functionality:
 - Periodically read the temperature sensor:
 - If the room is to warm ($>21^{\circ}\text{C}$) then open the air duct actuator to let in cold air.
 - If the rom is to cold ($\leq 21^{\circ}\text{C}$) then close the actuator.
 - Display relevant information on the LCD Screen.
 - Control the actuator.

Implementation and testing

Your implemented application code should be broken down (where possible) into discrete and efficient functions. The following is a list of basic main functional stages of the Microcontroller system (**NOTE:** This functionality can be changed according to reflect your design of the system, this is just a minimal list of functionalities):

STAGE 1 - Initialisation:

- Configure and enable GPIO port to read the onboard Pot and control the actuator.
- Configure and enable GPIO port for Controlling the onboard LCD.
- Configure interrupts if necessary.

STAGE 2 – Functional operation:

- Read the on board Pot which simulates a temperature sensor (i.e. read room temperature).
- Convert ADC reading to temperature. You can assume the full scale of the pot reads from 0°C to 100°C.
- Display Information on the LCD
- Perform actuator control.
 - Notes: The actuator should be at the 0° position when closed and 90° clockwise when open.
 - The actuator should take approx. 2 seconds to move from the open to closed position and approx. two seconds to move from the closed to open position.

Submission Deliverables:

1. Assignment source files for the implemented system (CCS Files), Assignment report and Demonstration Video should be submitted to blackboard by **11:59pm Friday 6th November 2026**).

Note: Source files, written report and demonstration video should be uploaded in a single zip file.

Marking Scheme

Written Report: (30%)

- A project report (max 3000 words – excluding cover page, table of contents, appendix etc), containing:
 1. Introduction: introduction to the project and proposed system including a problem statement and a statement of the functional requirements to be implemented.
 2. Design Choices: Review of the MSP430 and the hardware being used by the system with justification of the suitability of the hardware for the required system.
 3. System Design: A description of your system design process: e.g. , Methods/Functions required, Flow Charts, Schematics, Block Diagrams.
 4. Implementation and Testing: Briefly describe how the system was implemented at a high level then you should describe how you tested the system. E.g. Unit Testing and Integration Testing, Functional Testing and Debugging and Issue Resolution.

5. Conclusion: Summary of the project, achievements, future improvements and what was learned from the experience.

Practical Assessment (Code): (70%)

- Application file(s) (CCS project Files) that provides the functionality outlined in system requirements above.
- 5-10min demonstration video explaining how the system works, the implemented functionality and the testing applied to the system.
- Note: Code Should be commented
- **NOTE: These marks will be dependent on the complexity of your system solution and also how well structured your code is, i.e. code appropriately broken down into functions and the efficiency of design and implementation e.g. use of interrupts and timers.**
 - Stage 1 - Configuration of MCU
 - Stage 2 – Operation of MCU

MICR_ME601 Microcontrollers Mini Project – Written Reports Report Rubric (30%)

Criteria	Outstanding (80%-100%)	Excellent (70%-79%)	Good (60%-69%)	Acceptable (50%-59%)	Adequate (40%-49%)	Fail Limited (35%-39%)	Fail Unacceptable (0%-34%)
1. Introduction (Weighting: 2.5%)	Project and system overview are extremely clear, concise, and well contextualized. Problem statement and functional requirements are precise, complete, and highly relevant.	Clearly describes the project and its goals. Minor weaknesses in problem statement or requirement clarity.	Reasonably well-written; some areas may lack detail or clarity. Problem statement and requirements mostly aligned.	Adequate overview; lacks depth or clarity in some parts. Requirements are somewhat vague.	Weak overview and vague problem definition. Functional requirements are incomplete or misaligned.	Poor structure and unclear statements; minimal coverage of functional requirements.	No clear overview, problem definition, or requirements provided.
2. Design Choices: Review of MSP430 and Hardware Selection (Weighting: 5%)	Comprehensive and technically accurate discussion of MSP430 and hardware. Justification is strong and contextually relevant.	Very good understanding demonstrated. Minor gaps in technical details or justification.	Good explanation of MSP430 and hardware, though some components may lack sufficient depth or justification.	General overview is acceptable; some important points are missing or weakly explained.	Limited explanation with poor or unclear justification for hardware choices.	Minimal content provided; unclear or incorrect statements about components.	No evidence of research or understanding; critical technical inaccuracies.
3. System Design (Weighting: 7.5%)	Clear and coherent system description. All flowcharts, schematics, and block diagrams are accurate, complete, and easy to follow.	Well-structured design; most diagrams are correct and well-labelled. Minor clarity or detail issues.	Satisfactory system overview; diagrams included but may have errors or lack some clarity.	Acceptable level of system explanation; missing or unclear diagrams reduce effectiveness.	Poorly organized system design; diagrams are incomplete or poorly explained.	Inadequate structure and visuals; little evidence of system planning.	No meaningful system design or diagrams provided.
4. Implementation and Testing (Weighting: 10%)	Detailed methodology clearly outlines implementation and testing. Covers all testing types and shows excellent understanding of debugging.	Strong methodology and good coverage of testing strategies. Minor omissions or superficial debugging discussion.	Adequate implementation approach; testing is discussed but lacks depth.	Acceptable level of detail, but testing or debugging efforts are weak or underdeveloped.	Weak explanation of implementation; very limited testing strategies.	Testing and debugging are barely mentioned; unclear implementation process.	No implementation or testing approach described.
5. Conclusion (Weighting: 2.5%)	Insightful, well-structured summary of achievements and challenges. Excellent forward-looking suggestions.	Clear summary and good future recommendations; may miss minor points.	Acceptable recap with basic improvements noted. Some insight evident.	Summary lacks focus; future suggestions are underdeveloped.	Weak recap with vague or generic recommendations.	Very limited conclusion; no meaningful insights or future direction.	No conclusion or meaningful summary provided.
6. Report Structure and Presentation (Weighting: 2.5%)	Professionally formatted report with auto-generated ToC, clear headings,	Well-organized structure with minor formatting inconsistencies.	Report is readable with some formatting issues; visuals are present but may lack clarity.	Acceptable structure; some disorganization or missing elements like ToC.	Formatting and presentation hinder readability; figures may be mislabelled.	Poor presentation; many formatting issues and confusing layout.	Disorganized and unreadable; structure is missing or chaotic.

MICR_ME601 Microcontrollers Mini Project – Practical Assessment (70%)

	Outstanding	Excellent	Good	Acceptable	Adequate	Fail limited	Fail Unacceptable
	80 - 100%	70 - 79%	60 - 69%	50 - 59%	40 - 49%	21- 39%	0-20%
Practical Skills Assessment							
Microcontroller Configuration (Weighting: 15%)	Outstanding Configuration of the microcontroller. Masking techniques used everywhere appropriate and all configuration functionalised appropriately with appropriate bespoke header files.	Excellent Configuration of the microcontroller. Masking techniques used everywhere appropriate and Configuration broken into functions appropriately.	Good programming knowledge. Very good, correct configuration of the MCU. Appropriate masking techniques applied to the majority of the code. Configuration broken into functions appropriately in some places.	Acceptable evidence of programming knowledge. MCU mostly configured correctly. Appropriate masking techniques applied to the majority of the configuration.	Limited evidence of programming knowledge. MCU partially configured. Some appropriate masking techniques applied.	Little Evidence of programming Knowledge. MCU poorly configured. Inappropriate masking techniques used throughout.	Little or no evidence of programming knowledge required to implement the system. MCU not configured/ configured incorrectly. Inappropriate masking techniques used.
Microcontroller Operation (Weighting: 35%)	Extremely well structured and correct with all required functionality implemented correctly. Use of novel methods where appropriate. Outstanding code efficiency and test code used.	Extremely well structured and correct with all required functionality implemented correctly. Use of novel methods where appropriate. Excellent code efficiency and test code used.	Well structured and correct with all required functionality implemented correctly. Good code efficiency and test code used.	Well structured and mostly correct with the majority of required functionality implemented correctly. Reasonable thought given to code efficiency and test code used.	Well structured and mostly correct with the majority of required functionality implemented correctly. Little thought given to code efficiency and test code.	Little or no evidence of programming knowledge required to implement the system. System partially operates as per assignment brief.	Little or no evidence of programming knowledge required to implement the system. System fails to operate as per assignment brief
Demonstration Video Weighting (20%)	Extremely well presented and structured demonstration video. Clearly demonstrates all system functionality and how it was tested.	Excellent demonstration video that was very well thought out and presented. Demonstrates all functionality and shows a good amount of testing	Good demonstration of board functionality. Reasonably well structured and demonstrates the majority of functionality implemented. Shows some testing.	Good demonstration with good effort mad to demonstrate functionality and testing.	Reasonable demonstration but lacks real depth in demonstrating functionality and testing	Demonstration video provided but lacks knowledge of how the system should function and shows no testing.	Demonstration video missing or did not play.